Car Sales Potential Analysis in India Using Logistic Regression

1. Business Objective

The objective of this project is to analyse customer purchasing behaviour using the Japan automobile dataset and develop a classification model capable of predicting whether a customer is likely to purchase a new vehicle. The insights obtained from the Japanese market are then applied to the Indian customer dataset to estimate the potential market size and identify prospective customers.

The project aims to:

- Understand factors influencing customer purchase decisions.
- Build a predictive classification model using Logistic Regression.
- Evaluate the performance of the model using validation metrics.
- Estimate potential customers in the Indian market.
- Generate business insights through data visualization.

2. Dataset Description

Japan Dataset:

The Japanese dataset consists of 40,000 customer records as well as demographic and vehicle-related information.

Variables used in the analysis:

- Age Car Segment
- Age Group
- Income Group
- Gender Code
- Purchase (Target Variable)

India Dataset:

The Indian dataset contains 70,000 customer records.

Variables available:

- Age Car Segment
- Age Group
- Income Group
- Gender Code

Since the Purchase variable is unavailable in the Indian dataset, the trained model developed using the Japanese dataset was used to predict purchase behaviour.

3. Data Preparation

Before model development, the datasets were prepared and transformed.

Data Quality Checks:

- Missing values were checked and none were found.
- Duplicate records were verified and no duplicates were identified.
- Data types were validated before analysis.

Feature Engineering:

Age Group Segmentation:

Customers were categorized into age groups:

- Group 1: Below 30 years
- Group 2: 30–45 years
- Group 3: 45–60 years
- Group 4: Above 60 years

Age Car Segmentation:

Vehicle maintenance age was divided into four segments:

- Segment 1: Less than 200 days
- Segment 2: 200–360 days
- Segment 3: 361–500 days
- Segment 4: Greater than 500 days

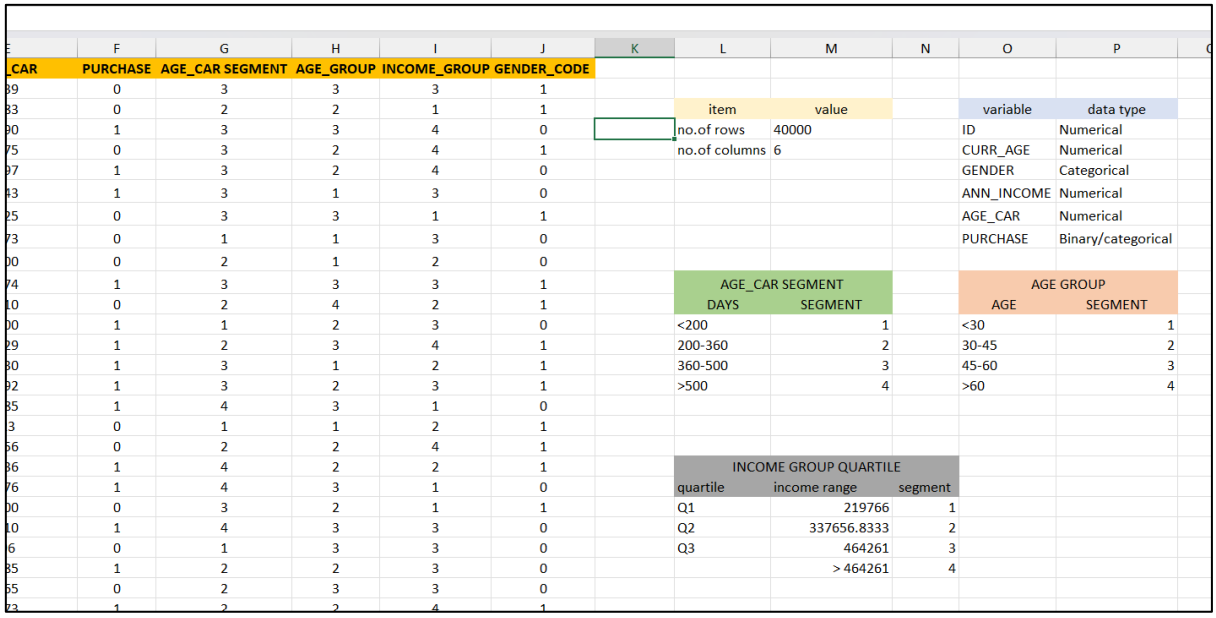
Income Group Segmentation:

Income was divided into quartiles:

- Group 1: Lowest income segment
- Group 2: Lower-middle income segment
- Group 3: Upper-middle income segment
- Group 4: Highest income segment

Gender Encoding:

- Male = 1
- Female = 0



_CAR	PURCHASE	AGE_CAR SEGMENT	AGE_GROUP	INCOME_GROUP	GENDER_CODE
89	0	3	3	3	1
83	0	2	2	1	1
90	1	3	3	4	0
75	0	3	2	4	1
97	1	3	2	4	0
43	1	3	1	3	0
25	0	3	3	1	1
73	0	1	1	3	0
00	0	2	1	2	0
74	1	3	3	3	1
10	0	2	4	2	1
00	1	1	2	3	0
29	1	2	3	4	1
80	1	3	1	2	1
92	1	3	2	3	1
85	1	4	3	1	0
3	0	1	1	2	1
56	0	2	2	4	1
86	1	4	2	2	1
76	1	4	3	1	0
00	0	3	2	1	1
10	1	4	3	3	0
6	0	1	3	3	0
85	1	2	2	3	0
55	0	2	3	3	0

item	value
no.of rows	40000
no.of columns	6

variable	data type
ID	Numerical
CURR_AGE	Numerical
GENDER	Categorical
ANN_INCOME	Numerical
AGE_CAR	Numerical
PURCHASE	Binary/categorical

AGE_CAR SEGMENT	
DAYS	SEGMENT
<200	1
200-360	2
360-500	3
>500	4

AGE GROUP	
AGE	SEGMENT
<30	1
30-45	2
45-60	3
>60	4

INCOME GROUP QUARTILE		
quartile	income range	segment
Q1	219766	1
Q2	337656.8333	2
Q3	464261	3
	> 464261	4

Figure 1: Feature engineering and variable transformation.

These transformations improved interpretability and aligned the variables with the project requirements.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer behaviour and identify factors influencing purchase decisions.

The Tableau dashboard included:

- Overall Purchase Distribution

- Purchase vs Gender
- Purchase vs Age Group
- Purchase vs Age Car Segment
- Purchase vs Income Group

Key Observations

1. Purchase distribution analysis showed that approximately 57.58% of customers purchased a new vehicle, while 42.42% did not.
2. Male customers demonstrated a slightly higher purchase tendency compared to female customers.
3. Customers belonging to higher income groups showed greater purchase rates, indicating that income plays an important role in purchasing decisions.
4. Customers with older vehicles were significantly more likely to purchase a new vehicle. The purchase rate increased from 36% in Segment 1 to 84% in Segment 4.
5. Age Groups 2 and 3 contributed the highest number of purchases, suggesting that middle-aged customers form an important target segment.

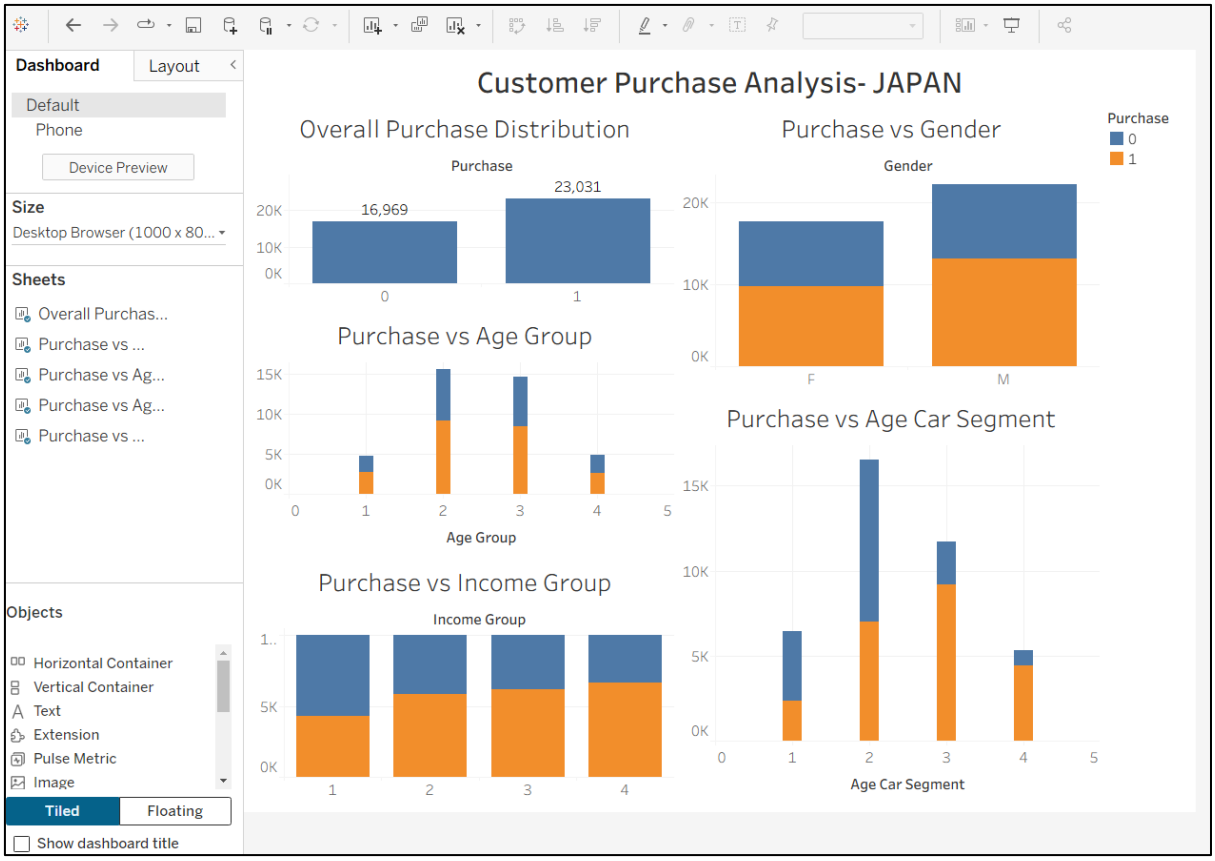


Figure 2: Tableau dashboard showing customer purchase patterns across gender, age group, income group, and age car segment.

These findings provided valuable business insights and supported the selection of variables used in the predictive model.

5. Model Building

A Logistic Regression model was developed using the Japan dataset.

Target Variable:

- Purchase

Independent Variables:

- Age Car Segment
- Age Group
- Income Group
- Gender Code

The dataset was divided into training and testing sets to evaluate model performance.

Justification for Variable Selection:

The variables Age Car Segment, Age Group, Income Group, and Gender Code were selected because exploratory analysis indicated a meaningful relationship between these variables and purchase behaviour.

- Age Car Segment was included because customers with older vehicles exhibited significantly higher purchase rates.
- Income Group was included because higher income levels were associated with increased purchasing activity.
- Age Group and Gender were included to capture demographic influences on customer decisions.

The model generated purchase probabilities for each customer.

A cut-off value of 0.5 was used:

- Probability $\geq 0.5 \rightarrow$ Purchase = 1
- Probability $< 0.5 \rightarrow$ Purchase = 0

	A	B	C	D	E	F	G
1	AGE_CAR SEGMENT	AGE_GROUP	INCOME_GROUP	GENDER_CODE	PURCHASE	PROBABILITY	PREDICTED
2	3	3	2	0	1	0.636494188	1
3	3	1	2	0	0	0.696984666	1
4	2	2	3	0	0	0.518122058	1
5	2	3	3	1	1	0.550455186	1
6	3	4	3	0	1	0.684005251	1
7	1	2	3	0	0	0.289050497	0
8	2	2	4	1	0	0.665381615	1
9	2	3	4	1	0	0.634360609	1
10	1	3	3	1	1	0.316476943	0
11	4	2	4	1	1	0.932918528	1
12	4	4	4	1	1	0.913695056	1
13	2	1	1	0	0	0.38036483	0
14	3	3	4	0	1	0.778526485	1
15	2	3	3	1	0	0.550455186	1
16	1	3	1	0	0	0.150163408	0
17	3	3	2	0	1	0.636494188	1
18	2	3	1	1	0	0.378855236	0
19	2	3	1	0	0	0.318472603	0
20	3	2	2	1	1	0.723716054	1
21	2	3	3	1	1	0.550455186	1
22	4	1	1	1	1	0.848570115	1
23	3	4	2	0	1	0.604387416	1
24	3	4	2	1	1	0.666004743	1
25	2	2	4	0	1	0.603717731	1
26	4	3	1	1	1	0.810095373	1
27	3	3	1	1	0	0.61730148	1
28	2	2	2	1	0	0.497612518	0

Figure 3: Logistic Regression model implementation and probability prediction.

6. Model Evaluation

The model was evaluated using the testing dataset.

Confusion Matrix:

The confusion matrix was used to compare actual and predicted outcomes and assess classification performance.

	PREDICTED	
ACTUAL	1	0
1	5306	2885
0	2118	1691

Figure 4: Confusion Matrix used for model evaluation.

Validation Metrics:

Metric	Value
Accuracy	58.31%
Precision	71.47%
Recall	64.78%
F1 Score	67.96%
Cut-off Value	0.50
AUC	0.7465

METRIC	VALUE
ACCURACY	58.31%
PRECISION	71.47%
RECALL	64.78%
F1-SCORE	67.96%
CUT-OFF	0.5
AUC	0.7465

Figure 5: Model performance metrics.

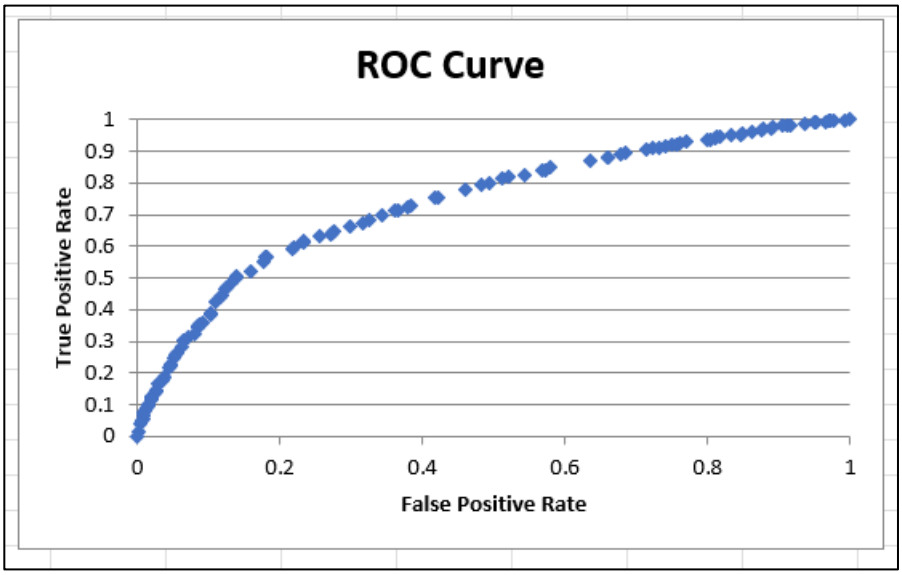


Figure 6: ROC Curve showing model discrimination capability.

These metrics indicate that the model demonstrates reasonable predictive capability and can be used to estimate customer purchase potential.

7. Business Interpretation of Model Coefficients

The Logistic Regression coefficients provide insights into how different variables influence purchase probability.

- Age Car Segment emerged as the strongest positive predictor of purchase behaviour. Customers whose vehicles have not undergone recent maintenance are more likely to purchase a new vehicle.
- Income Group showed a positive relationship with purchase probability, indicating that customers with higher incomes have greater purchasing capability.
- Gender demonstrated a moderate positive influence on purchase behaviour.
- Age Group showed a slight negative relationship with purchase probability, suggesting that older customer groups may be somewhat less likely to purchase compared to younger and middle-aged customers.

Logistic Regression							
		# Iter	20		Alpha	0.05	
	coeff	*s.e.*	*Wald*	*p-value*	*exp(b)*	*lower*	*upper*
intercept	-2.6451	0.06409	1703.29	0	0.071		
AGE_CAR SEGMENT	0.97252	0.01649	3479.11	0	2.64461	2.56051	2.73146
AGE_GROUP	-0.1364	0.01565	75.9267	0	0.87252	0.84617	0.8997
INCOME_GROUP	0.34846	0.01233	798.176	0	1.41688	1.38304	1.45155
GENDER_CODE	0.26639	0.02684	98.488	0	1.30525	1.23835	1.37576

Figure 7: Logistic Regression coefficient estimates.

These insights can help businesses design targeted marketing strategies and improve customer acquisition efforts.

8. Prediction on India Dataset

- The trained Logistic Regression model was applied to the Indian customer dataset containing 70,000 records.
- Purchase probabilities were calculated for each customer using the coefficients obtained from the Japanese dataset.
- Customers with probability greater than or equal to the cut-off value of 0.5 were classified as potential buyers.

Prediction Results:

- Total Customers Analysed: 70,000
- Predicted Potential Customers: 41,344
- Potential Customer Percentage: 59.06%

L	M
PROBABILITY	PREDICTED_PURCHASE
0.76405301	1
0.82105183	1
0.85803813	1
0.82240101	1
0.16841253	0
0.39835117	0
0.4651715	0
0.80115041	1
0.92386163	1
0.5188211	1
0.18740969	0
0.88262847	1
0.15016341	0
0.24748704	0
0.26184521	0
0.91419919	1
0.20907171	0
0.15016341	0
0.2002296	0
0.84145579	1
0.84022324	1
0.78822013	1
0.71272178	1
0.38036483	0
0.33449149	0

Figure 8: Prediction and Probability results on the Indian customer dataset.

Potential Customers	Percentage
41344	59.06%

Figure 9: Potential customers results on the Indian customer dataset.

This result represents the estimated market potential identified within the available Indian dataset.

9. Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Focus marketing campaigns on higher-income customer segments.
- Prioritize customers owning older vehicles, as they demonstrate higher purchase intent.
- Target Age Groups 2 and 3, which contribute significantly to purchase activity.
- Use predictive analytics to identify high-potential customers and optimize marketing expenditure.

These strategies can improve customer targeting and increase conversion rates.

10. Conclusion

A Logistic Regression model was successfully developed using the Japan dataset and applied to the Indian dataset to estimate customer purchase potential.

The analysis identified Age Car Segment, Income Group, Age Group, and Gender as important factors influencing customer purchase behaviour. The model achieved acceptable predictive performance and successfully estimated market potential in India.

Out of 70,000 customers analysed, 41,344 customers were identified as potential buyers, representing 59.06% of the dataset. The findings demonstrate how customer demographics and vehicle-related characteristics can be leveraged to support data-driven business decisions and market expansion strategies.